Project – 5 Household Electricity Consumption Prediction

Project Title

Household Electricity Consumption Prediction Using Traditional Machine Learning

1. Real-Life Problem

Electricity providers and smart-home systems need to understand and predict electricity consumption. Accurate predictions can support demand planning, energy management and identification of unusual consumption patterns.

2. Problem Statement

> Develop a machine-learning regression system capable of predicting household electricity consumption using historical electrical measurements and temporal information.

3. Dataset

UCI Individual Household Electric Power Consumption Dataset

This is a substantial real-world time-series dataset containing **2,075,259 measurements**, collected at one-minute intervals over almost four years from a household in France. It contains nine electrical measurements and includes missing values in approximately 1.25% of rows.

UCI Individual Household Electric Power Consumption Dataset:

https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

Dataset download link:

https://archive.ics.uci.edu/static/public/235/individual+household+electric+power+consumption.zip

4. ML Task

Time-series regression

Possible target:

`Global_active_power`

5. Important Features

- Global active power
- Global reactive power
- Voltage
- Global intensity
- Sub-metering 1
- Sub-metering 2
- Sub-metering 3

- Date
- Time

6. Objectives

Students should:

1. Process the large time-series dataset.
2. Handle missing observations.
3. Perform temporal EDA.
4. Analyze daily/weekly/seasonal consumption patterns.
5. Create temporal features.
6. Engineer lag features.
7. Engineer rolling statistical features.
8. Develop regression models.
9. Investigate lag-based prediction.
10. Investigate feature importance.
11. Analyze prediction errors.

7. Suggested Algorithms

- Linear Regression
- Ridge
- Lasso
- Decision Tree
- Random Forest
- Gradient Boosting
- Extra Trees
- SVR

8. Feature Engineering

Students should investigate:

Temporal features

- Hour
- Minute
- Day
- Day of week
- Month
- Weekend

Lag features

- Previous 1 minute
- Previous 10 minutes
- Previous 30 minutes
- Previous 1 hour
- Previous 24 hours

Rolling features

- Rolling mean
- Rolling maximum
- Rolling minimum
- Rolling standard deviation

9. Evaluation Metrics

- MAE
- RMSE
- MAPE

10. Important Experimental Requirement

Students **must use chronological train/validation/test splitting**.

They should **not randomly shuffle the complete dataset before splitting**, because that can allow information from the future to influence the model-development process.

11. Capstone Investigation

How does prediction performance change as the forecasting horizon increases?

For example:

Experiment	Prediction
E1	Next 10 minutes
E2	Next 30 minutes
E3	Next 1 hour
E4	Next 6 hours
E5	Next 24 hours

Students can then investigate how traditional ML models behave as the prediction horizon increases.

12. Expected Deliverables

- Large-scale data preprocessing
- Time-series EDA
- Feature-engineering
- Multiple regression models
- Forecast-horizon experiment
- Error analysis
- Feature importance
- Final prediction system

13. Specific Tasks

1. Load the large dataset efficiently.
2. Examine data types and missing values.
3. Convert date/time information appropriately.

4. Analyze electricity consumption distribution.
5. Analyze hourly consumption.
6. Analyze daily consumption.
7. Analyze weekly patterns.
8. Create temporal features.
9. Create lag features.
10. Create rolling statistical features.
11. Develop Linear Regression.
12. Develop Ridge/Lasso.
13. Develop Decision Tree.
14. Develop Random Forest.
15. Develop Gradient Boosting.
16. Compare different forecasting horizons.

Teams may perform additional experiments without introducing deep learning.

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14. Possible Experiments

Experiment A — Baseline

Predict using the immediately available electrical variables.

Experiment B — Temporal Features

Add:

- Hour
- Day
- Day of week
- Month
- Weekend

Experiment C — Lag Features

Investigate:

- Previous observation
- Previous 10-minute value
- Previous 30-minute value
- Previous-hour value
- Previous-day value

Experiment D — Rolling Statistics

Investigate:

- Rolling mean
- Rolling standard deviation
- Rolling minimum

- Rolling maximum

Experiment E — Forecast Horizon

Compare prediction performance at different horizons, for example:

Experiment	Horizon
E1	10 minutes
E2	30 minutes
E3	1 hour
E4	6 hours
E5	24 hours

Important: Every experiment must use only information that would genuinely be available at the prediction time.

15. Expected Graphs

1. Global active power distribution
2. Power consumption over time
3. Hourly consumption pattern
4. Daily consumption pattern
5. Weekly consumption pattern
6. Monthly consumption pattern
7. Correlation heatmap
8. Lag correlation plot
9. Rolling mean plot
10. Actual vs predicted consumption
11. Residual distribution
12. Forecast-horizon vs RMSE
13. Feature importance
14. Model comparison